

Christian Venturella

Theoretical Chemist - Yale University, New Haven, CT

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website - github - LinkedIn

INSTITUTIONS

Ph.D. Student, Theoretical Chemistry, Yale University

SEP 2022 — CURRENT

Bachelors in Chemistry, Princeton University

SEP 2017 — MAY 2021

Minors: Applied and Computational Mathematics, Applications of Computing, Materials Science and Engineering

AWARDS AND HONORS

Graduate Awards

NDSEG Fellowship (2024-Current), NSF GRFP Fellowship Awardee (2024), Kirkwood Fellowship (2024-Current), ACCESS Discover Allocation Principal Investigator (2026)

Undergraduate Awards

summa cum laude (2021), American Institute of Chemists Student Award (2021), Sigma Xi Research Society (2021), Fulbright Semifinalist (2021)

RESEARCH AND WORK EXPERIENCE

Ph.D. Research | Theoretical Chemistry | Yale University

AUG 2022 — CURRENT

- Machine Learning Many-Body Hamiltonian and Green's Function
 - Formulated machine learning (ML) method for predicting many-body quantum chemistry via Green's function (MBGF) and self-energy in a symmetry and polarization adapted basis
 - Developed graph attention network architectures for predicting many-body physics by targeting the self-energy over DFT orbital graphs.
 - Generated and maintained databases of DFT, GW , coupled cluster, simulations for ML training and validation
 - Formulated ML method for predicting DFT Hamiltonian from low-cost orbital descriptors
- Unsupervised Learning of Chromophore Design Space from Ab Initio Excited State Descriptors
 - (ACCESS PI 2026) Constructed comprehensive database of 200,000 chromophore structures from chemical literature and calculated their optical and magnetic response properties with TDDFT
 - Formulated low-rank GGA response kernel with Gaussian density-fitting for high efficiency and throughput
 - Developed generative AI models to generate new molecules with desirable properties
- Low Scaling Green's Functions Calculations of Molecules via Interacting Bath Dynamical Embedding Theory
 - Formulated low-cost GW and EOM-CCSD methods for molecules based on quantum embedding
 - Implemented low rank algorithms for constructing cluster-specific natural orbitals in the bath expansion
 - Benchmarked accuracy of predicted charged excitation energies with ORCA and PySCF
- Enzyme Reaction Mechanism Elucidation (Experimental Collaboration)
 - Collaborated with organic chemists to realize their proposed catalytic pathways in DFT simulations
 - Sampled reactive binding modes with semi-empirical methods and classical dynamics
 - Predicted free energy changes and barrier heights for multistep routes

Undergraduate Research | Computational Chemistry | Princeton University

SEP 2020 — APR 2021

- Correlated enzyme efficiency to substrate binding free energy with molecular dynamics simulations
- Investigated light-driven reaction mechanism with time-dependent density functional theory

Software Engineering and Data Science | Educational Data Analytics | LinkIt, NYC

JUL 2021 — JUL 2022

- Engineered a data analysis system for generation of regular administrative reports in PDF and web formats from database views
- Automated process of computing statewide educational metrics for 500+ school districts
- Designed calibrated statistical models for pinpointing students at risk of not meeting educational standards

PUBLICATIONS

Low-Scaling Many-Body Green's Function Calculations for Molecular Systems via ibDET	APR 2026
Venturella, C.; Li, J.; Zhu, T. 2026, arXiv:2604.0313	
Unified Deep Learning Framework for Many-Body Quantum Chemistry via Green's Functions	JUN 2025
Venturella, C.; Li, J.; Hillenbrand, C.; Peralta, X. L.; Liu, J.; Zhu, T. Nat. Comput. Sci. 2025, 5, 502–513, arXiv:2407.20384	
Machine learning many-body Green's function for molecular excitation spectra	OCT 2023
C. Venturella, C. Hillenbrand, J. Li, and T. Zhu, J. Chem. Theory Comput. 2024, 20, 1, 143–154, arXiv:2310.09911	
Modeling Nonnatural Flavoenzyme Catalysis with Molecular Dynamics and Quantum Chemical Methods	2021
Princeton Undergraduate Thesis, Advisor(s): Gregory Scholes, Daniel Oblinsky	

PRESENTATIONS

American Chemical Society National Meeting, Atlanta, GA	Spring 2026
American Physical Society Global Physics Summit, Anaheim, CA	Spring 2025
Computational Molecular Sciences & Engineering Symposium, Yale University (Symposium Link)	Fall 2024
Yale Chemistry Symposium, Yale University	Summer 2024
American Chemical Society National Meeting, New Orleans	Spring 2024

TEACHING AND MENTORSHIP

Graduate Research Mentor, Zhu Group	SEP 2023 — Current
<i>Yale University Chemistry Department</i>	
• Mentor multiple senior undergraduates in research projects • Work with Yale University STARS program to support underrepresented groups in STEM • Built interactive software demonstrations and organized Yale Chemistry Department open house for middle and high school students	
Teaching Fellow	SEP 2022 — Current
<i>Yale University Chemistry Department</i>	
• General Chemistry I	SEP 2022 — JUN 2023
• Physical Chemistry I (Thermodynamics and Kinetics)	SEP 2023 — DEC 2023
Undergraduate Teaching Assistant (Organic Chemistry I/II)	SEP 2020 — MAY 2021
<i>Princeton University Chemistry Department</i>	
Trenton Arts Program Volunteer	SEP 2018 — MAY 2021
<i>Princeton University Music Department</i>	

SKILLS

Computational Chemistry

Python for chemical modeling and informatics (pyscf, geometric, ASE, rdkit), cp2k, Gaussian, xTB

Data Science and Data Engineering

Python for ML (scipy, sklearn, PyTorch, PyTorch Geometric, TensorFlow), python for data analytics and visualization (pandas, matplotlib, plotly, xlwings, dash, cvx, geopandas), SQL, MATLAB, R, Microsoft Excel

Mathematical Modeling

Kernel Ridge Regression, Gaussian Process Regression, Neural Networks, Geometric Deep Learning, Convex Optimization, Classifier Models, Model Calibration

Word Processing

L^AT_EX, HTML, Microsoft Word

Languages

English (Native), Spanish (Intermediate)